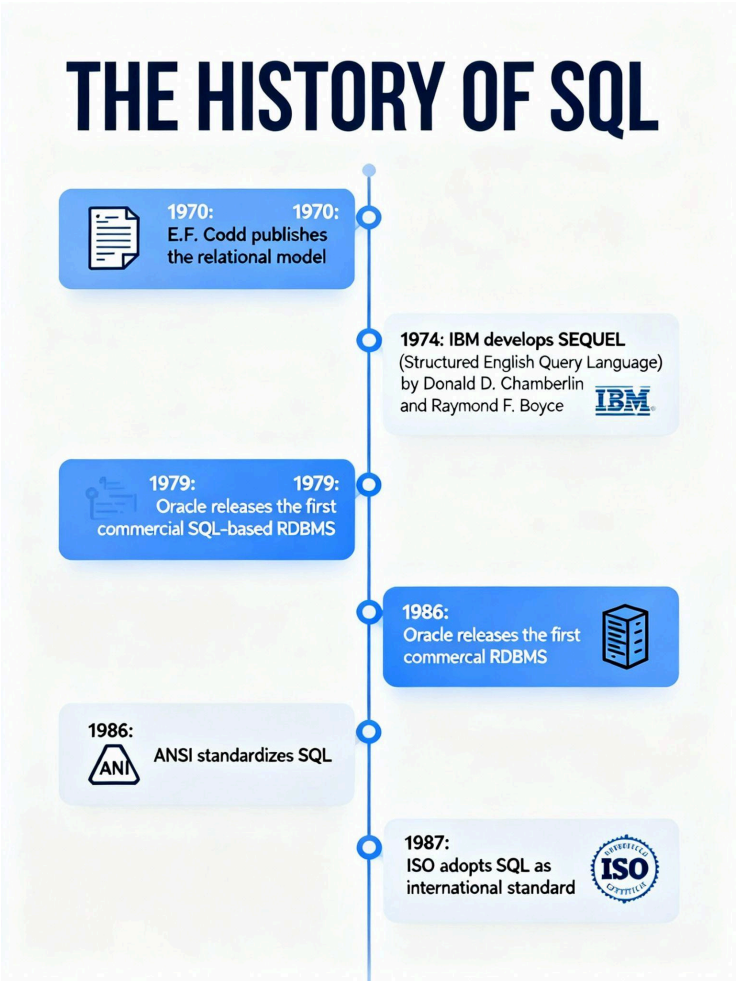
Database Management Systems – Day 3: SQL Overview

1. History of SQL

- **SQL (Structured Query Language)** was developed in the **1970s** by IBM researchers **Donald D. Chamberlin** and **Raymond F. Boyce**.
- It was initially called **SEQUEL (Structured English Query Language)**.
- Adopted by the **American National Standards Institute (ANSI)** in 1986 and the **International Organization for Standardization (ISO)** in 1987.
- SQL became the **standard language for relational database management systems (RDBMS).**

Timeline Highlights:

Year	Event
1970	E.F. Codd publishes the relational model
1974	IBM develops SEQUEL
1979	Oracle releases the first commercial SQL-based RDBMS
1986	ANSI standardizes SQL
1987	ISO adopts SQL as international standard



💡 **Key Point:** SQL is the universal language for interacting with relational databases like MySQL, Oracle, SQL Server, and PostgreSQL.

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2. Features of SQL

- **High-level language:** SQL allows users to query and manipulate data without knowing the underlying database implementation.
- **Non-procedural:** Focuses on *what to do*, not *how to do it*.
- **Standardized:** Follows ANSI/ISO standards for compatibility across RDBMS.
- **Powerful:** Can handle complex queries, joins, transactions, and aggregations.
- **Secure:** Supports access control using DCL commands (GRANT, REVOKE).

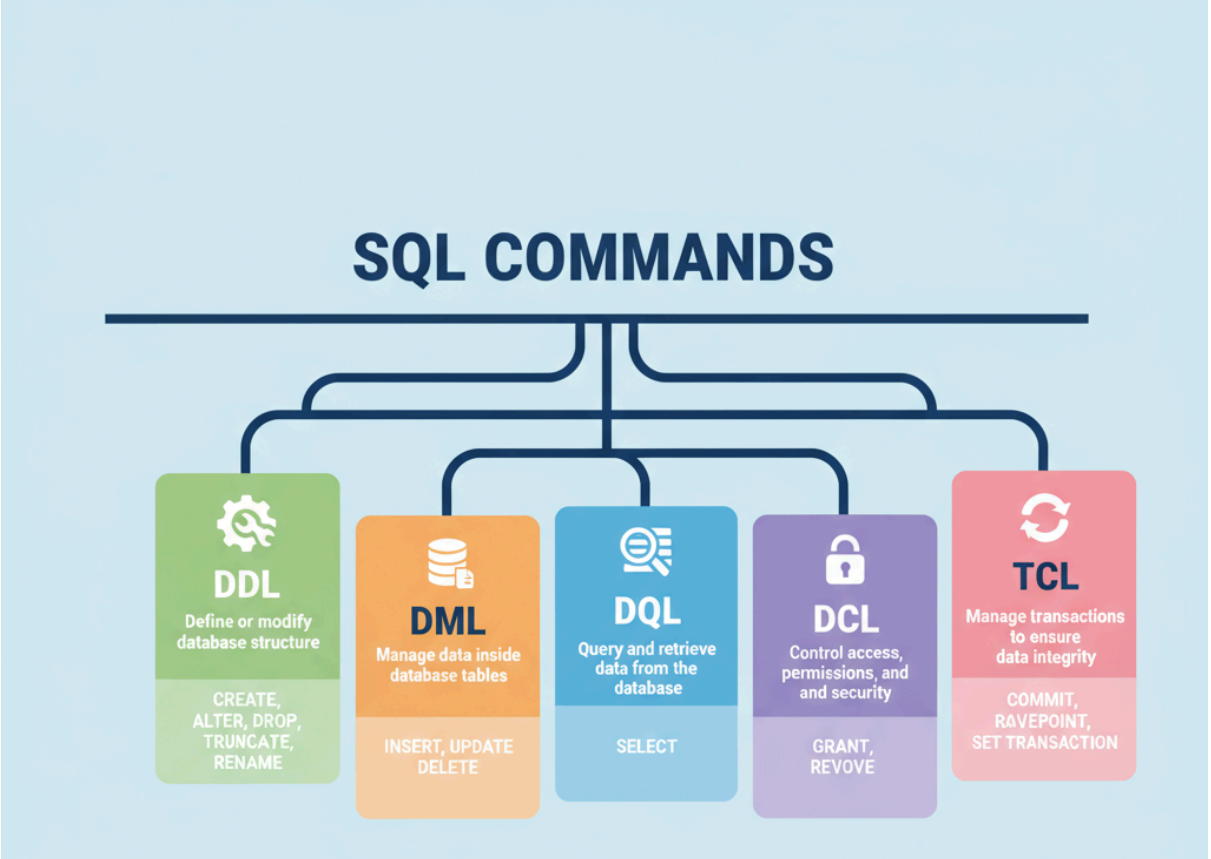
- **Multi-user:** Allows concurrent access with ACID transaction support.

Example: Retrieve all students above age 20:

```
SELECT Name, Age FROM Students WHERE Age > 20;
```

3. Types of SQL Commands

SQL commands are categorized based on functionality into **five major groups**:



3.1 Data Definition Language (DDL)

- **Purpose:** Define or modify database structure (tables, schemas, indexes).

- **Commands:**

Command	Description	Example
CREATE	Creates database or table	`CREATE TABLE Students (ID INT, Name VARCHAR(50));`
ALTER	Modifies existing table	`ALTER TABLE Students ADD COLUMN Age INT;`
DROP	Deletes table or database	`DROP TABLE Students;`
TRUNCATE	Removes all records but keeps table	`TRUNCATE TABLE Students;`
RENAME	Renames table or column	`ALTER TABLE Students RENAME TO Learners;`

> ✅ **Effect:** DDL commands **change the structure** of the database. Changes are **auto-committed**.

3.2 Data Manipulation Language (DML)

- **Purpose:** Manage data inside database tables (insert, update, delete).
- **Commands:**

Command	Description	Example
INSERT	Add new records	`INSERT INTO Students VALUES (1, 'Alice', 20);`
UPDATE	Modify existing records	`UPDATE Students SET Age = 21 WHERE Name='Alice';`
DELETE	Remove records	`DELETE FROM Students WHERE ID=1;`

> ✅ **Effect:** DML commands **manipulate data** but do not alter table structure. Changes can be **committed or rolled back**.

3.3 Data Query Language (DQL)

- **Purpose:** Query and retrieve data from the database.
- **Main Command:**

Command	Description	Example
SELECT	Fetch records from one or more tables	`SELECT Name, Age FROM Students WHERE Age > 18;`

- **Clauses for SELECT:**
  - `WHERE` – Filter records
  - `GROUP BY` – Aggregate records
  - `ORDER BY` – Sort results
  - `JOIN` – Combine multiple tables

💡 **Tip:** DQL is read-only; it does not modify data.

3.4 Data Control Language (DCL)

- **Purpose:** Control access, permissions, and security.
- **Commands:**

Command	Description	Example
GRANT	Give privileges to users	`GRANT SELECT, INSERT ON Students TO 'user1';`
REVOKE	Remove privileges	`REVOKE INSERT ON Students FROM 'user1';`

✅ **Effect:** DCL ensures **security and controlled access** in multi-user environments.

3.5 Transaction Control Language (TCL)

- **Purpose:** Manage transactions to ensure data integrity.
- **Commands:**

Command	Description	Example
COMMIT	Save all changes permanently	`COMMIT;`
ROLLBACK	Undo changes since last commit	`ROLLBACK;`
SAVEPOINT	Set a point to rollback partially	`SAVEPOINT sp1;`
SET TRANSACTION	Define transaction properties	`SET TRANSACTION READ ONLY;`

> ✅ **Effect:** TCL ensures **ACID properties** and maintains database consistency during operations.

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4. Summary Diagram

```
SQL Commands
├── DDL    : CREATE, ALTER, DROP, TRUNCATE, RENAME
├── DML    : INSERT, UPDATE, DELETE
├── DQL    : SELECT
├── DCL    : GRANT, REVOKE
└── TCL    : COMMIT, ROLLBACK, SAVEPOINT, SET TRANSACTION
```

> SQL is the **backbone of relational databases**, allowing creation, manipulation, querying, security, and transaction control.

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5. Example: SQL Workflow

1. **Create a table (DDL)**

```
CREATE TABLE Students (
    ID INT PRIMARY KEY,
```

```
    Name VARCHAR(50),
    Age INT
);
```

2. **Insert records (DML)**

```
INSERT INTO Students VALUES (1, 'Alice', 20);
INSERT INTO Students VALUES (2, 'Bob', 22);
```

3. **Query records (DQL)**

```
SELECT * FROM Students;
```

4. **Grant access (DCL)**

```
GRANT SELECT ON Students TO 'user1';
```

5. **Commit transaction (TCL)**

```
COMMIT;
```